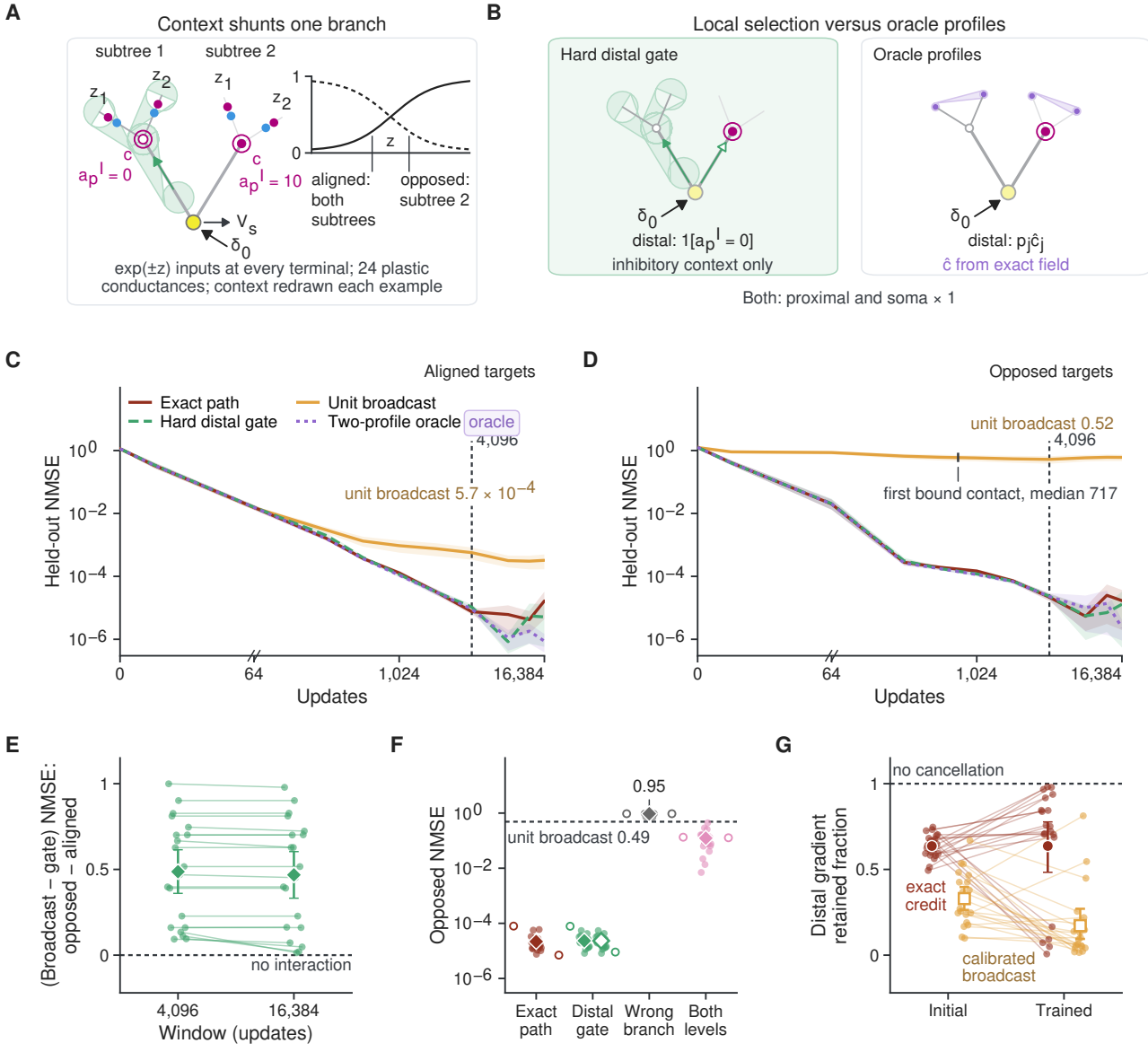

A
Context shunts one branch
subtree 1
subtree 2
z1
z2
c
a_p^I = 0
a_p^I = 10
V_s
δ0
aligned: both subtrees
opposed: subtree 2
exp(±z) inputs at every terminal; 24 plastic conductances; context redrawn each example
B
Local selection versus oracle profiles
Hard distal gate
distal: 1[a_p^I = 0]
inhibitory context only
Oracle profiles
distal: p_j ĉ_j
ĉ from exact field
Both: proximal and soma × 1
C
Aligned targets
Exact path
Hard distal gate
Unit broadcast
Two-profile oracle
oracle
4,096
unit broadcast 5.7 × 10−4
Held-out NMSE
Updates
D
Opposed targets
unit broadcast 0.52
4,096
first bound contact, median 717
Held-out NMSE
Updates
E
(Broadcast − gate) NMSE: opposed − aligned
no interaction
4,096
16,384
Window (updates)
F
Opposed NMSE
0.95
unit broadcast 0.49
Exact path
Distal gate
Wrong branch
Both levels
G
Distal gradient retained fraction
no cancellation
exact credit
calibrated broadcast
Initial
Trained